JIAZHENG ZHU

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Research Profile

I am a PHD student with broad interests in **time-domain astrophysics and optical transients**. My research combines wide-field transient surveys with follow-up observations and multi-wavelength data analysis, with particular expertise in **tidal disruption events (TDEs)**. I have experience across different stages of transient research, from survey observations and transient discovery to photometric and spectroscopic follow-up and physical interpretation. Looking toward the era of increasingly deep time-domain surveys, particularly the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST), I am especially interested in exploring the faint and still poorly sampled regimes of the transient sky. My future research will focus on using TDEs to probe black holes across new mass and redshift regimes, particularly **TDEs from intermediate-mass black holes (IMBHs)** and **high-redshift TDEs**. By combining next-generation surveys with rapid multi-wavelength follow-up and spectroscopy, I aim to uncover transient populations that have remained largely inaccessible to previous surveys and use them to investigate black-hole demographics and transient accretion physics.

Research Interests

- **Time-domain Astrophysics & Optical Transients:** Discovery and physical characterization of optical transients using wide-field surveys, photometric and spectroscopic follow-up, and multi-wavelength observations.
- **Tidal Disruption Events:** Observational diversity, transient accretion physics, host-galaxy environments, and the use of TDEs as probes of otherwise quiescent black holes.
- **High-redshift TDEs:** Extending TDE discovery and spectroscopy toward higher redshift to study transient black-hole accretion and TDE populations across cosmic time.
- **Intermediate-mass Black Holes:** Searching for TDEs and other transient signatures of intermediate-mass black holes in low-mass galaxies and other environments where dormant black holes are difficult to identify.
- **Faint & Unexplored Transient Populations:** Using increasingly deep time-domain surveys to uncover transient populations that have been poorly sampled by previous surveys, including faint nuclear transients and unusual accretion events.

Education

- **University of Science and Technology of China (USTC)** — *2021 – Present*
  Ph.D. Candidate in Astrophysics, School of Physical Sciences — Hefei, China
  - Ph.D. program through an integrated Master's–Doctoral track.
  - Advisors: Prof. Tinggui Wang and Prof. Ning Jiang.

- **University of Science and Technology of China (USTC)** — *2017 – 2021*
  B.S. in Astronomy, School of Physical Sciences — Hefei, China

Publications

4 first-author publications and 18 co-authored publications

First-author Publications

[1] **Zhu, J.**, Xu, Z., Jiang, N., et al. (2026). **Discovery of a Featureless Tidal Disruption Event at $z \sim 1$ with the Wide Field Survey Telescope**. *The Astrophysical Journal Letters*, 1004, L10. DOI: 10.3847/2041-8213/ae710d.

[2] **Zhu, J.**, Jiang, N., Wang, Y., et al. (2025). **Ultraviolet Spectral Evidence for Ansky as a Slowly Evolving Featureless Tidal Disruption Event with Quasiperiodic Eruptions**. *The Astrophysical Journal Letters*, 994, L16. DOI: 10.3847/2041-8213/ae19ea.

[3] **Zhu, J.**, Jiang, N., Wang, T., et al. (2023). **AT 2023clx: The Faintest and Closest Optical Tidal Disruption Event Discovered in Nearby Star-forming Galaxy NGC 3799**. *The Astrophysical Journal Letters*, 952, L35. DOI: 10.3847/2041-8213/ace625.

[4] **Zhu, J.**, Jiang, N., Dong, S., et al. (2023). **SN 2017egm: A Helium-rich Superluminous Supernova with Multiple Bumps in the Light Curves**. *The Astrophysical Journal*, 949, 23. DOI: 10.3847/1538-4357/acc2c3.

Co-author Publications

[5] Liu, H., Sun, L., Jiang, N., Shu, X., Wang, Y., Wang, T., Cutri, R. M., Dou, L., Zhang, F., **Zhu, J.**, et al. (2026). **An Obscured Tidal Disruption Event Uncovered by Its Mid- and Near-Infrared Dust Echo in a Star-forming Galaxy**. *The Astrophysical Journal*, 1005, 178. DOI: 10.3847/1538-4357/ae7362.

[6] Li, D., Zhang, W., Yang, J., Chen, J.-H., Yuan, W., Cheng, H., Xu, F., Shu, X., Shen, R.-F., Jiang, N., **Zhu, J.**, et al. (2026). **A Fast Powerful X-ray Transient from Possible Tidal Disruption of a White Dwarf**. *Science Bulletin*, 71, 538–546. DOI: 10.1016/j.scib.2025.12.050.

[7] Wang, J., Huang, M., Xue, Y., Jiang, N., Huang, S., Wang, Y., **Zhu, J.**, et al. (2026). **A Tidal Disruption Event from an Intermediate-mass Black Hole Revealed by Comprehensive Multi-wavelength Observations**. *Nature Communications*, 17, 2007. DOI: 10.1038/s41467-026-68670-3.

[8] Wu, M., Jiang, N., **Zhu, J.**, Luo, D., Dou, L., Wang, T. (2025). **A Torus Remnant Revealed by the Infrared Echo of Tidal Disruption Event AT 2019qiz: Implications for the Missing Energy and Quasiperiodic Eruption Formation**. *The Astrophysical Journal Letters*, 988, L77. DOI: 10.3847/2041-8213/adf229.

[9] Huang, S., Wang, T., Jiang, N., Shen, R.-F., Chen, Z., Wang, Y., **Zhu, J.**, et al. (2025). **Unveiling the Cosmic Dance of Repeated Nuclear Transient ASASSN-14ko: Insights from Multiwavelength Observations**. *The Astrophysical Journal*, 988, 237. DOI: 10.3847/1538-4357/ade795.

[10] Lin, J., Wang, T., Cai, M., et al. (2025). **Minute-cadence Observations of the Galactic Plane with the Wide Field Survey Telescope (WFST): Overview, Methodology, and Early Results**. *The Astrophysical Journal Supplement Series*, 278, 29. DOI: 10.3847/1538-4365/adc0fc.

[11] Jiang, N., Luo, D., **Zhu, J.**, Cutri, R. M. (2025). **The Extraordinary Long-lasting Infrared Echo of PS16dtm Reveals an Extremely Energetic Nuclear Outburst**. *The Astrophysical Journal Letters*, 980, L17. DOI: 10.3847/2041-8213/adaeb9.

[12] Liu, Y., Fan, L., Hu, L., Lu, J., Lu, Y., Xu, Z., **Zhu, J.**, et al. (2025). **Classification of Real and Bogus Transients Using Active Learning and Semi-supervised Learning**. *Astronomy & Astrophysics*, 693, A105. DOI: 10.1051/0004-6361/202348581.

[13] Jin, C.-C., Li, D.-Y., Jiang, N., Dai, L.-X., Cheng, H.-Q., **Zhu, J.**, et al. (2025). **An Intermediate-mass Black Hole Lurking in a Galactic Halo Caught Alive during Outburst**. *arXiv e-prints*, arXiv:2501.09580. DOI: 10.48550/arXiv.2501.09580.

[14] Sun, L., Jiang, N., Dou, L., Shu, X., **Zhu, J.**, et al. (2024). **Recurring Tidal Disruption Events a Decade Apart in IRAS F01004-2237**. *Astronomy & Astrophysics*, 692, A262. DOI: 10.1051/0004-6361/202452380.

[15] Lin, Z., Jiang, N., Wang, T., Kong, X., Li, D., He, H., **Zhu, J.**, et al. (2024). **The Unluckiest Star: A Spectroscopically Confirmed Repeated Partial Tidal Disruption Event AT 2022dbl**. *The Astrophysical Journal Letters*, 971, L26. DOI: 10.3847/2041-8213/ad638e.

[16] Wang, Y., Wang, T., Jiang, N., Zhang, X., **Zhu, J.**, et al. (2024). **ASASSN-18ap: A Dusty Tidal Disruption Event Candidate with an Early Bump in the Light Curve**. *The Astrophysical Journal*, 966, 136. DOI: 10.3847/1538-4357/ad2ae4.

[17] Huang, S., Jiang, N., Lin, Z., **Zhu, J.**, Wang, T. (2024). **Correction to: AT2018dyk Revisited: A Tidal Disruption Event Candidate with Prominent Infrared Echo and Delayed X-ray Emission in a LINER Galaxy**. *Monthly Notices of the Royal Astronomical Society*, 531, 4793. DOI: 10.1093/mnras/stae1445.

[18] Huang, S., Jiang, N., **Zhu, J.**, Wang, Y., Wang, T., et al. (2024). **AT 2023lli: A Tidal Disruption Event with Prominent Optical Early Bump and Delayed Episodic X-Ray Emission**. *The Astrophysical Journal Letters*, 964, L22. DOI: 10.3847/2041-8213/ad319f.

[19] Jiang, N., Zhou, Z., **Zhu, J.**, Wang, Y., Wang, T. (2023). **Two Candidate Obscured Tidal Disruption Events Coincident with High-energy Neutrinos**. *The Astrophysical Journal Letters*, 953, L12. DOI: 10.3847/2041-8213/acebe3.

[20] Lin, Z., Jiang, N., Kong, X., Huang, S., Lin, Z., **Zhu, J.**, et al. (2022). **The Luminosity Function of Tidal Disruption Flares for the ZTF-I Survey**. *The Astrophysical Journal Letters*, 939, L33. DOI: 10.3847/2041-8213/ac9c63.

[21] Dou, L., Jiang, N., Wang, T., Shu, X., Yang, H., Pan, Z., **Zhu, J.**, et al. (2022). **X-ray View of a Merging Supermassive Black Hole Binary Candidate SDSS J1430+2303: Results from the First ∼200 Days of Observations**. *Astronomy & Astrophysics*, 665, L3. DOI: 10.1051/0004-6361/202244450.

[22] Wang, Y., Jiang, N., Wang, T., **Zhu, J.**, et al. (2022). **Discovery of ATLAS17jrp as an Optical-, X-Ray-, and Infrared-bright Tidal Disruption Event in a Star-forming Galaxy**. *The Astrophysical Journal Letters*, 930, L4. DOI: 10.3847/2041-8213/ac6670.

Observational Experience & Telescope Time

Observing Experience

- **Wide Field Survey Telescope (WFST), 2023–Present.** Participated in observing and survey operations, with a cumulative one week on site and six weeks remotely. Responsibilities included observing assistance, monitoring observing conditions, and real-time image-quality assessment.
- **Lijiang 2.4-m Telescope, 2024.** Completed two nights of on-site observing.

Principal Investigator (PI) Programs

- **Gemini North, 2024B.** "Towards the first optical discovery of an IMBH-TDE at its Eddington accretion phase in a dwarf starburst galaxy." Program ID: **GN-2024B-DD-105**. Approved time: **2 hours**.
- **Swift UVOT/XRT, 2023–2026.** Transient target-of-opportunity requests submitted in response to individual sources. Total approved time: **129 ks**.

Co-Investigator (Co-I) Programs

- **JWST.** "Are LFBOTs Dusty? Constraining the Nature of IR Emission in the Luminous Fast Blue Optical Transient AT2026dbl." Program ID: **JWST-DD-12569**. Approved time: **5.2 hours**.
- **Gemini North, 2024A.** "Towards a Deep Spectroscopic View of the First X-Ray IMBH-TDE Candidate Discovered at Early Phase." Program ID: **GN-2024A-DD-103**. PI: Ning Jiang. Approved time: **3 hours**.
- **Gemini North, 2023A.** "Spectroscopic Monitoring of a SMBH Binary Candidate at its Last Inspiraling Stage." Program ID: **GN-2023A-Q-110**. PI: Ning Jiang. Approved time: **4.5 hours**.
- **Gemini North, 2022A.** "Spectroscopic Diagnosis of Optical flares from a merging supermassive black hole binary." Program ID: **GN-2022A-DD-101**. PI: Zhen Pan. Approved time: **2 hours**.
- **HST.** "Imaging of EP241021a during the flattening in the optical light curve: an exotic gamma-ray burst or a peculiar jetted tidal disruption event?" Program ID: **HST-GO/DD-18213**. Approved time: **2 orbits**.
- **HST.** "Turn-on AGN or IMBH-TDE? UV Spectroscopic Observations May Tell." Program ID: **HST-GO/DD-17933**. Approved time: **4 orbits**.
- **HST.** "Probing the UV spectral evolution of a unique TDE with early bright X-ray emission." Program ID: **HST-GO/DD-17849**. Approved time: **4 orbits**.
- **HST.** "A UV spectroscopic view of the 2024 outburst of AT2020afhd: A Unique Opportunity to Test the Reprocessing Model of Tidal Disruption Event." Program ID: **HST-GO/DD-17609**. Approved time: **2 orbits**.
- **VLA, 2026A.** "VLA Observation of a Unique Off-nuclear TDE Candidate." Program ID: **VLA/26A-638**. Proposal type: Director's Discretionary Time. Proposal date: 2 June 2026. Approved time: **2 hours at Priority B**, plus **8 hours at Priority C** conditional on detection in the initial observation.
- **LCOGT, 2023A.** "Continuous Daily Optical Monitoring of a SMBH Binary Candidate at its Last Inspiraling Stage." Program ID: **NAOC2023A-005**. PI: Ning Jiang. Approved time: **35 hours**.
- **LCOGT, 2022B.** "Further Daily Optical Monitoring of an unprecedented SMBH Binary at its Last Inspiraling Stage." Program ID: **NAOC2022B-002**. PI: Ning Jiang. Approved time: **20 hours**.
- **MMT/Binospec, 2025B.** "Spectroscopic Identification of Candidate Tidal Disruption Events Caused by Intermediate-Mass Black Holes." Program ID: **UAO-S111-25B**. PI: Ann Zabludoff. Approved time: **1.5 nights** of queue observing.

Additional collaborative observing proposals include P200, Swift, XMM-Newton, and the Lijiang 2.4-m telescope; program details are not listed here.

Research & Technical Skills

- **Time-domain Surveys & Transient Discovery:** Transient detection and classification, survey data mining, light-curve analysis, variability characterization, multi-band photometry, and candidate selection
- **Astronomical Imaging & Photometry:** Raw image processing, CCD calibration, image stacking, astrometry, image subtraction, difference imaging, and photometric measurements
- **Spectroscopy:** Optical/UV spectroscopic analysis, continuum characterization, emission-line analysis, and transient classification
- **Transient SED & Physical Modeling:** Multi-wavelength spectral energy distribution (SED) fitting, blackbody and continuum modeling, MCMC, and Bayesian inference
- **Host-galaxy Analysis:** Galaxy photometry and morphology, stellar-population analysis, host-galaxy SED fitting, and spectral fitting
- **Programming & Scientific Computing:** Programming, scientific computing, astronomical data analysis, and visualization, primarily using Python

Honors & Awards

- **National Scholarship for Graduate Students** *2023*
  Ministry of Education of the People's Republic of China
  ◦ Awarded the National Scholarship for outstanding academic and research performance.

MEETING

- **Tidal Disruption Events and Nuclear Transients: Entering the Data-Rich Era** *2024*
  Heraklion, Crete, Greece on 2-6 September 2024
  - Mid-Infrared Outbursts in Nearby Galaxies(MIRONG) – oral

- **Tidal Disruption Events and Nuclear Transients: Entering the Data-Rich Era** *2024*
  Heraklion, Crete, Greece on 2-6 September 2024
  - AT2023clx: A new representative of whole faint TDE population – poster